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Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Not specified in the provided report.

Diagnosis: Invasive ductal carcinoma, grade 3 with ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: The patient, age 32, diagnosed with breast cancer. ER-positive, PR-positive, HER2-negative.
- Generation IV:

Summary:

National Comprehensive Cancer Network® (NCCN®)

Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Met, as the patient was diagnosed at age 32., Triple-negative breast cancer: Not met, as the patient's cancer is ER-positive and PR-positive., Multiple primary breast cancers: Not indicated., Lobular breast cancer with personal or family history of diffuse gastric cancer: Not applicable., Male breast cancer: Not applicable., Ancestry: Ashkenazi Jewish ancestry: Not indicated., At least 1 close blood relative with breast,

ovarian, pancreatic, prostate cancer, or other specified criteria: Not applicable based on the provided information.

Conclusion

The patient meets the NCCN criteria for genetic testing based on the diagnosis of breast cancer before age 50. There is no family history or hereditary indicators provided in the pedigree, but the early age of onset warrants consideration for testing for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2.

Medical Report

Pathology Report Summary

Clinical History: Patient diagnosed with breast cancer at age 32.

Diagnosis: Invasive ductal carcinoma, grade 3; Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: The patient, diagnosed with breast cancer at age 32.
- Generation IV:

Textual Analysis:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 years: Met, as the patient was diagnosed at age 32., Triple-negative breast cancer: Not met, as the patient is ER-positive and PR-positive.

Conclusion

The patient is eligible for NCCN testing criteria due to the diagnosis of breast cancer at age 32, well below the age of 50.

Medical Report

Pathology Report Summary

Case Number: 2

Age of Diagnosis: 28

Diagnosis:

- Invasive Ductal Carcinoma: High-grade invasive ductal carcinoma, grade 3.
- Ductal Carcinoma in Situ (DCIS): High grade with comedo necrosis.

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: The patient, a female, diagnosed with invasive ductal carcinoma at age 28., Patient's sister diagnosed with ovarian cancer at age 38.
- Generation IV:

Image Description:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years., Any age with ≥ 1 close blood relative with ovarian cancer.

Conclusion

The patient is eligible for NCCN criteria testing.

Due to the patient's diagnosis of breast cancer at a young age (28) and a family history of a sister with ovarian cancer, she qualifies for genetic testing according to the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). Recommended testing for high-penetrance breast cancer susceptibility genes includes BRCA1 and BRCA2, among others.

Medical Report

Pathology Report Summary

Clinical History: Case of invasive ductal carcinoma, grade 3.

Specimen: Not specified.

Diagnosis: Invasive ductal carcinoma, grade 3; Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient: Female, diagnosed with breast cancer at age 28, ER-positive, PR-positive, HER2-positive., Sister: Diagnosed with ovarian cancer at age 38.
- Generation IV: Children, Nieces, Nephews

Image Link:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at age 28., Close blood relative (sister) with ovarian cancer.

Conclusion

The patient is eligible for NCCN testing criteria based on her diagnosis of breast cancer at age 28 and a sister's history of ovarian cancer. Genetic testing for high-penetrance breast cancer susceptibility genes is recommended.

Medical Report

Pathology Report Summary

Case Number: 3

Patient Age at Diagnosis: 52 years

Diagnosis:

Invasive ductal carcinoma, Grade 3 (high - grade invasive ductal carcinoma)

Ductal carcinoma in situ (DCIS), high grade with comedo necrosis

Hormone Receptor Status:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2: Positive

Pedigree Analysis

The patient's pedigree, based on the provided information, shows no family history of breast, ovarian, pancreatic, or prostate cancers.

There are no documented relatives with early - onset breast cancer (i.e., at or below age 50), no male breast cancer in the family, and no mention of other syndromic cancers (e.g., gastric cancer).

NCCN Testing Criteria Assessment

Age at diagnosis (52) is above the ≤ 50 threshold.

The tumor is HER2 - positive (thus not triple - negative).

Only one breast cancer (invasive plus adjacent DCIS is generally considered a single primary).

There is no family history of the relevant cancer.

There is no mention of Ashkenazi Jewish ancestry or other risk features.

Conclusion

Based on the above findings, the patient does not meet any of the NCCN testing criteria for high - penetrance breast cancer susceptibility genes. Therefore, she is not currently eligible for NCCN - recommended genetic testing under these guidelines.

Medical Report

Pathology Report Summary

Case Number: 3

Patient's Age at Diagnosis: 52 years

Pathological Findings:

Invasive ductal carcinoma (Grade 3; high-grade)

High-grade ductal carcinoma in situ (DCIS) with comedo necrosis

Hormone Receptor Profile:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Positive

Pedigree Analysis

The patient's pedigree indicates no family history of breast, ovarian, pancreatic, or prostate cancer. No relatives are documented with early-onset breast cancer (≤ 50 years), no male breast cancer in the family, and no other notable syndromic cancers (e.g., diffuse gastric cancer).

At diagnosis, the patient was 52 years old (above the ≤ 50 threshold).

The tumor is HER2-positive, so it is not triple-negative.

There is a single breast cancer (invasive + DCIS generally considered one primary).

No family history of the applicable cancers (breast ≤ 50 , ovarian, male breast, pancreatic, or high-risk prostate), and no mention of Ashkenazi Jewish ancestry.

NCCN Testing Criteria Assessment

Age at diagnosis (52) is above the ≤ 50 threshold.

The tumor is HER2-positive (thus not triple-negative).

Only one breast cancer (invasive plus adjacent DCIS is generally considered a single primary).

There is no family history of the relevant cancer.

There is no mention of Ashkenazi Jewish ancestry or other risk features.

Conclusion

Since none of the NCCN criteria apply, this patient does not qualify for genetic testing for high-penetrance breast cancer genes. She is not eligible according to current NCCN Guidelines®

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Specimen details not provided.

Diagnosis: Invasive ductal carcinoma, grade 3: High-grade invasive ductal carcinoma. Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2/neu: Negative

Note: High-grade invasive ductal carcinoma with positive ER and PR status.

Pedigree Analysis

Generations:

- Generation I: Maternal Grandmother: breast cancer at age 42.
- Generation II: Parents
- Generation III: The patient, a female, diagnosed with breast cancer at age 55, ER-positive, PR-positive, and HER2-negative.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at 55, which does not meet this criterion., Triple-negative breast cancer: The patient's cancer is not triple-negative (ER and PR positive)., Family history of breast cancer: Maternal grandmother had breast cancer at age 42, fulfilling the criterion of a close blood relative with breast cancer at age ≤ 50 .

Conclusion

The patient qualifies for NCCN testing due to having a maternal grandmother diagnosed with breast cancer at age 42, which meets the guideline of a close blood relative with breast cancer at age ≤ 50 . Genetic testing for high-penetrance breast cancer susceptibility genes (such as BRCA1 and BRCA2) is recommended based on this family history.

Medical Report

Pathology Report Summary

Case Number: 4

Diagnosis Age: 55

Diagnosis: Invasive ductal carcinoma, grade 3. High-grade ductal carcinoma in situ (DCIS) with comedo necrosis.

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Maternal grandmother with breast cancer at age 42.
- Generation II: Parents
- Generation III: The patient diagnosed with breast cancer at age 55.

Image Description:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 y: Not met (diagnosed at 55)., Any age with ≥ 1 close blood relative with breast cancer at age ≤ 50 : Met (maternal grandmother diagnosed at 42).

Conclusion

The patient is eligible for NCCN testing criteria due to a family history of breast cancer with the maternal grandmother diagnosed at a young age (42). Genetic testing for high-penetrance breast cancer susceptibility genes is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, diagnosed at age 39.

Specimen: Not specified

Diagnosis: Invasive ductal carcinoma, grade 3 with ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2/neu: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Father had pancreatic cancer diagnosed at age 60.
- Generation III: The patient, diagnosed with high-grade invasive ductal carcinoma at age 39.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Met, diagnosed at age 39., Triple-negative breast cancer: Not met, as HER2 is positive., Close blood relative with pancreatic cancer: Met, father had pancreatic cancer at age 60.

Conclusion

The patient is eligible for NCCN genetic testing criteria, with a personal history of breast cancer diagnosed at age 39 and a family history of a close blood relative (father) with pancreatic cancer.

Medical Report

Pathology Report Summary

Clinical History: A 39-year-old diagnosed with breast cancer.

Specimen: Breast tissue sample.

Diagnosis: Invasive ductal carcinoma, grade 3, with high-grade ductal carcinoma in situ and comedo necrosis.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Father diagnosed with pancreatic cancer at age 60.
- Generation III: The patient diagnosed with high-grade invasive ductal carcinoma at age 39, ER-negative, PR-negative, HER2-positive.
- Generation IV: Children, Nieces, Nephews

Visual:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: Patient diagnosed at 39. Any age with Triple-negative breast cancer: ER-negative, PR-negative, HER2-positive is considered HER2-positive subtype, not triple-negative. Any age with ≥ 1 close blood relative with pancreatic cancer: Father diagnosed with pancreatic cancer.

Conclusion

The patient meets the NCCN testing criteria for BRCA1, BRCA2, and other high-penetrance breast cancer susceptibility genes due to being diagnosed with breast cancer at age 39 and having a close blood relative with pancreatic cancer. Testing is strongly recommended.

Medical Report

Pathology Report Summary

Case Number: 6

Age of Diagnosis: 53

Family History: No family history

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: Patient diagnosed with breast cancer at age 53, ER-positive, PR-positive, HER2-positive.
- Generation IV: Children, no documented cases of cancer.

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 y: Not met, since diagnosed at 53., Triple-negative breast cancer: Not applicable, patient is ER, PR, HER2 positive., Multiple primary breast cancers: Not applicable., Lobular breast cancer with personal/family history of diffuse gastric cancer: Not applicable., Male breast cancer: Not applicable., Ashkenazi Jewish ancestry: Not applicable., Close blood relative with breast cancer at age ≤ 50 : Not applicable., Close blood relative with male breast cancer: Not applicable., Close blood relative with ovarian, pancreatic, or prostate cancer: Not applicable.

Conclusion

The patient does not meet the NCCN criteria for testing for high-penetrance breast cancer susceptibility genes. There is no reported family history or personal history that aligns with the NCCN testing guidelines.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Not specified.

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Parents
- Generation III: Patient (Case 6), diagnosed with breast cancer at age 53.
- Generation IV: Children

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer: Diagnosed at age 53, not meeting the age ≤ 50 criterion., Triple-positive breast cancer (ER, PR, and HER2 positive): Does not meet the triple-negative breast cancer criterion., No family history: No additional criteria met from family history.

Conclusion

The patient does not meet the NCCN testing criteria for BRCA1, BRCA2, and other high-penetrance breast cancer susceptibility genes due to the age of diagnosis being over 50 and the absence of relevant family history or other specific criteria listed. Consultation for genetic testing does not appear warranted based on these guidelines.

Medical Report

Pathology Report Summary

Case Number: 7

Age of Diagnosis: 62

Family History: No family history

Diagnosis: Invasive breast carcinoma of no special type, grade 3. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Grandparents
- Generation II: Parents
- Generation III: Patient (Affected)

Textual Analysis:

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer.

Conclusion

The patient, diagnosed with triple-negative breast cancer at age 62, does meet the NCCN testing criteria for genetic testing based on having triple-negative breast cancer. Despite the absence of a family history of related cancers, the triple-negative status alone qualifies the patient for further genetic testing under the guidelines.

Medical Report

Pathology Report Summary

Case Number: 7

Age of Diagnosis: 62

Diagnosis: Invasive breast carcinoma of no special type, grade 3. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Family History: No family history

Pedigree Analysis

Generations:

- Generation I: No relevant medical history noted.
- Generation II: Parents
- Generation III: The patient, diagnosed at 62 with triple-negative breast cancer.
- Generation IV: Children, no noted diagnosis.

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer

Criteria Not Met: Personal history of breast cancer with age ≤ 50 y., Any age with Multiple primary breast cancers., Any age with Lobular breast cancer with personal or family history of diffuse gastric cancer., Any age with Male breast cancer., Any age with Ashkenazi Jewish ancestry., Any age with ≥ 1 close blood relative with breast cancer at age ≤ 50 ., Any age with ≥ 1 close blood relative with male breast cancer., Any age with ≥ 1 close blood relative with ovarian cancer., Any age with ≥ 1 close blood relative with pancreatic cancer., Any age with ≥ 1 close blood relative with prostate cancer with metastatic, or high- or very-high-risk group., Any age with ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family.

Conclusion

The patient is eligible for the NCCN testing criteria based on the diagnosis of triple-negative breast cancer. Despite the lack of family history of cancer, the triple-negative status itself qualifies the patient for genetic testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2, etc.

Medical Report

Pathology Report Summary

Clinical History: Case of breast cancer in a 35-year-old female.

Specimen: Not specified in the provided document.

Diagnosis: Invasive breast carcinoma of no special type, grade 2, with intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: The patient's mother had breast cancer, diagnosed at age 45.
- Generation III: Case 8, diagnosed with breast cancer at age 35., ER-positive, PR-positive, HER2-negative
- Generation IV: Children and other descendants not detailed.

ImageLink:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Met, as the patient was diagnosed at 35., Family history of breast cancer: Met, as the patient's mother had breast cancer at age 45.

Conclusion

The patient meets the NCCN criteria for genetic testing based on her early age of breast cancer diagnosis and her family history, specifically her mother's diagnosis at a relatively young age. Testing for high-penetrance breast cancer susceptibility genes is recommended, including BRCA1, BRCA2, and others mentioned in the criteria.

Medical Report

Pathology Report Summary

Clinical History: A case of invasive breast carcinoma for review.

Specimen: Breast tissue biopsy.

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Mother diagnosed with breast cancer at age 45.
- Generation III: The patient, diagnosed with breast cancer at age 35, ER-positive, PR-positive, and HER2-negative.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at 35, meeting this criterion., Family history of breast cancer: The patient's mother was diagnosed at age 45, further supporting the consideration.

Conclusion

The patient meets the NCCN criteria for genetic testing based on the diagnosis of breast cancer at age 35 and family history of breast cancer in the mother diagnosed at age 45. Genetic testing for high-penetrance breast cancer susceptibility genes like BRCA1, BRCA2 may be recommended.

Medical Report

Pathology Report Summary

Case Number: 9

Age of Diagnosis: 54

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Uncle
- Generation II: Aunt, Uncle, Paternal Cousin with Male Breast Cancer
- Generation III: Patient: Diagnosed with breast cancer at age 54, ER-negative, PR-negative, HER2-positive.

Family History:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 is not met as patient was diagnosed at 54., Any age with Triple-negative breast cancer is not met as ER and PR are negative, but HER2 is positive., Any age with Male breast cancer: Met due to the patient's paternal cousin having male breast cancer.

Conclusion

The patient is eligible for NCCN testing based on the family history of a paternal cousin with male breast cancer.

Although the patient does not meet the criteria for young age at diagnosis or triple-negative breast cancer, the presence of male breast cancer in a close relative (paternal cousin) justifies genetic testing for high-penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Case Number: 9

Age of Diagnosis: 54

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Aunts/Uncles
- Generation II: Aunts, Uncles
- Generation III: Patient diagnosed with breast cancer at age 54. Paternal cousin male diagnosed with breast cancer at age 50.

NCCN Testing Criteria Assessment

Criteria Met: Any age with ≥ 1 close blood relative with male breast cancer.

Conclusion

The patient meets the NCCN testing criteria due to a close blood relative (paternal cousin) with male breast cancer. Therefore, the patient is eligible for testing for high-penetrance breast cancer susceptibility genes, particularly BRCA1 and BRCA2.

Medical Report

Pathology Report Summary

Clinical History: A case of invasive breast carcinoma, for review.

Case Number: 10

Diagnosis: Invasive breast carcinoma of no special type, grade 3. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Age of Diagnosis: 40

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient diagnosed with invasive breast carcinoma at age 40, ER-positive, PR-negative, HER2-positive., Maternal Aunt with ovarian cancer diagnosed at age 50.
- Generation IV: Children, Nieces, Nephews

Pedigree Image:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years: Diagnosed at age 40., Family history of ovarian cancer: Maternal aunt diagnosed at age 50.

Conclusion

The patient meets the NCCN Guidelines for genetic testing due to a diagnosis of breast cancer at age 40, which is under the age threshold of 50, as well as a family history of ovarian cancer in a close relative (maternal aunt). Therefore, the patient is eligible for testing of high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2, among others. Genetic counseling is recommended to discuss the potential implications of test results and guide further family screening and management.

Medical Report

Pathology Report Summary

Case Number: 10

Age of Diagnosis: 40

Diagnosis: Invasive breast carcinoma of no special type, grade 3. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Maternal Aunt: Ovarian cancer at age 50.
- Generation II:
- Generation III: Patient: Breast cancer diagnosed at age 40.

Textual Analysis:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years., Close blood relative (maternal aunt) with ovarian cancer.

Conclusion

The patient is eligible for NCCN testing criteria. The diagnosis of breast cancer at age 40 and a maternal aunt with ovarian cancer at age 50 meet the guidelines for testing for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Case Number: 11

Age of Diagnosis: 70

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Negative
- HER2: Negative

Family History: No family history reported.

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Affected Patient, diagnosed with breast cancer at age 70, ER positive, PR negative, HER2 negative, Siblings, First Cousins
- Generation IV: Children, Nieces, Nephews

Analysis:

NCCN Testing Criteria Assessment

Criteria: Age at diagnosis is 70, which does not meet the criterion of ≤ 50 years., The cancer is not triple-negative., No report of multiple primary breast cancers., No lobular breast cancer or family history of diffuse gastric cancer mentioned., Patient is female; criterion for male breast cancer not applicable., No information on Ashkenazi Jewish ancestry is provided., No close blood relatives with breast, ovarian, pancreatic, or prostate cancer reported.

Conclusion

The patient does not meet the NCCN criteria for genetic testing based on the available information. There is no significant family history or specific tumor characteristics that align with the testing guidelines for high-penetrance breast cancer susceptibility genes. Further testing is not indicated unless additional risk factors are identified.

Medical Report

Pathology Report Summary

Clinical History: Case of invasive breast carcinoma diagnosed at age 70.

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Negative
- HER2: Negative

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient (diagnosed with breast cancer at age 70)
- Generation IV: Children, Nieces, Nephews

Image Description:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years is not met (diagnosed at 70)., Triple-negative breast cancer is not met (ER positive, PR negative, HER2 negative)., Multiple primary breast cancers not noted., No family history of diffuse gastric cancer or male breast cancer., No Ashkenazi Jewish ancestry mentioned., No close blood relatives with breast cancer at age ≤ 50 years., No close blood relatives with ovarian, pancreatic, or high-risk prostate cancer., No multiple diagnoses on the same side of the family.

Conclusion

The patient does not meet the NCCN testing criteria for high-penetrance breast cancer susceptibility genes based on age at diagnosis, receptor status, and family history. Given the lack of family history and the patient's receptor status, genetic testing is not indicated based on NCCN Guidelines.

Medical Report

Pathology Report Summary

Case Number: 12

Age of Diagnosis: 38

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Parents
- Generation III: Patient: Case 12, Breast cancer diagnosed at 38, ER-negative, PR-positive, HER2-negative., Brother: Metastatic prostate cancer diagnosed at age 45.
- Generation IV: Children, Nieces, Nephews

Description:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer diagnosed at age ≤ 50 years: Met, diagnosed at age 38., Family history of prostate cancer with metastatic condition: Met, brother diagnosed with metastatic prostate cancer at age 45.

Conclusion

The patient meets the NCCN testing criteria due to her personal history of breast cancer diagnosed at an age ≤ 50 (38 years) and a family history of metastatic prostate cancer in her brother. Genetic testing for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2, is recommended based on this history. Further familial genetic counseling and testing may be advised.

Medical Report

Pathology Report Summary

Case Number: 12

Age of Diagnosis: 38

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: Patient: Case 12, diagnosed with breast cancer at age 38, ER-negative, PR-positive, HER2-negative., Brother: Diagnosed with metastatic prostate cancer at age 45.
- Generation IV:

Visual Representation:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at age 38., Close blood relative with prostate cancer with metastatic risk: The patient's brother had metastatic prostate cancer at age 45.

Conclusion

The patient is eligible for NCCN high-penetrance breast cancer susceptibility gene testing. Factors include diagnosis of breast cancer at age 38 and a brother with metastatic prostate cancer at age 45. Testing for BRCA1, BRCA2, and other relevant genes is recommended based on these criteria.

Medical Report

Pathology Report Summary

Case Number: 13

Age of Diagnosis: 29

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Negative
- HER2: Negative

Pedigree Analysis

Generations:

- Generation II: Patient: Female diagnosed with breast cancer at age 29. ER-positive, PR-negative, HER2-negative., Sister: Diagnosed with triple-negative breast cancer at age 34.

Visual Representation:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at age 29, which meets this criterion., Family history: Sister diagnosed with triple-negative breast cancer at age 34, which meets the criterion of having a close blood relative with breast cancer at age ≤ 50 .

Conclusion

The patient is eligible for NCCN testing.

The patient meets multiple criteria for NCCN testing, including a personal history of breast cancer diagnosed at a young age (29 years) and a family history of early-onset breast cancer in a sibling. These factors warrant genetic testing for high-penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer for review.

Age at Diagnosis: 29

Diagnosis: Invasive breast carcinoma of no special type, grade 2 and intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Negative
- HER2: Negative

Note: Standard hormone-based treatment may be considered due to ER-positive status.

Pedigree Analysis

Generations:

- Generation II: The patient diagnosed with breast cancer at age 29., Sister diagnosed with triple-negative breast cancer at age 34.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 y: The patient was diagnosed at age 29., Family history of triple-negative breast cancer: Sister diagnosed at age 34.

Conclusion

The patient meets the NCCN testing criteria due to a diagnosis of breast cancer at age 29 and a family history of triple-negative breast cancer in a sister diagnosed at age 34. Genetic testing for high-penetrance breast cancer susceptibility genes, such as BRCA1 and BRCA2, is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of invasive breast carcinoma for review.

Age of Diagnosis: 62

Diagnosis: Invasive breast carcinoma of no special type, grade 2.

Ductal Carcinoma In Situ: Intermediate grade with a solid pattern.

Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Siblings, Half-siblings, First cousins
- Generation IV: Children, Nieces, Nephews

Patient Family:

NCCN Testing Criteria Assessment

Criteria: Patient age at diagnosis is 62, which does not meet the criteria for age ≤ 50 ., Hormone receptor positive status (Triple-negative criterion not met)., No reported family members with breast cancer or other related cancers included in criteria.

Conclusion

The patient does not meet the NCCN testing criteria for high-penetrance breast cancer susceptibility genes. There is no family history of related cancers, and the diagnosis was made at an age older than 50 with hormone receptor positive status.

Medical Report

Pathology Report Summary

Clinical History: Case of breast cancer, evaluation requested.

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Uncles
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient (Case 14): Diagnosed with invasive breast carcinoma at age 62 - ER-positive, PR-positive, HER2-negative.
- Generation IV: Children, Nieces, Nephews

Pedigree Image:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 : Not met (diagnosed at 62)., Triple-negative breast cancer: Not met (ER-positive, PR-positive, HER2-negative)., Multiple primary breast cancers: Not applicable., Lobular breast cancer with personal or family history of diffuse gastric cancer: Not applicable., Male breast cancer: Not applicable., Ashkenazi Jewish ancestry: Not noted., Close blood relative with breast cancer at age ≤ 50 : Not noted., Close blood relative with male breast cancer: Not noted., Close blood relative with ovarian cancer: Not noted., Close blood relative with pancreatic cancer: Not noted., Close blood relative with prostate cancer: Not noted., Three diagnoses of breast and/or prostate cancer on the same side of the family: Not applicable.

Conclusion

The patient does not meet the NCCN criteria for high-penetrance breast cancer gene testing. There is no significant family history or personal history that aligns with the specific conditions set forth for genetic testing under the listed criteria.

Medical Report

Pathology Report Summary

Case Number: 15

Age of Diagnosis: 46

Diagnosis:

- Primary Diagnosis: Invasive breast carcinoma of no special type, grade 1
- Secondary Diagnosis: Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Father, pancreatic cancer at age 50, Paternal uncle, pancreatic cancer at age 50
- Generation II: Patient, breast cancer at age 46

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: Patient was diagnosed at 46., Any age with ≥ 1 close blood relative with pancreatic cancer: Father and paternal uncle both had pancreatic cancer.

Conclusion

The patient is eligible for NCCN testing.

The patient meets multiple NCCN criteria for genetic testing, including having breast cancer diagnosed at or before age 50 and having close relatives with pancreatic cancer. Testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2, and others is recommended.

Medical Report

Pathology Report Summary

Case Number: 15

Clinical History: Diagnosis of invasive breast carcinoma at age 46.

Diagnosis:

- Invasive Breast Carcinoma: No special type, grade 1.
- Ductal Carcinoma In Situ: Intermediate grade with a solid pattern.

Hormone Receptor Status:

- Estrogen Receptor (ER): Negative
- Progesterone Receptor (PR): Negative
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Patient's father diagnosed with pancreatic cancer at age 50., Paternal uncle diagnosed with pancreatic cancer at age 50.
- Generation III: Patient (Case 15) diagnosed with breast cancer at age 46.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 y, Family history of pancreatic cancer (father and uncle).

Conclusion

The patient is eligible for NCCN testing criteria due to the history of being diagnosed with breast cancer at age 46 and having a family history of pancreatic cancer.

Medical Report

Pathology Report Summary

Case Number: 16

Age of Diagnosis: 28

Diagnosis: Invasive breast carcinoma of no special type, grade 2 and Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation II: The patient's mother had breast cancer diagnosed at age 48.
- Generation III: The patient, a female, diagnosed with breast cancer at age 28.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 y., Any age with Triple-negative breast cancer., Any age with ≥ 1 close blood relative with breast cancer at age ≤ 50 .

Conclusion

The patient meets the NCCN testing criteria due to a personal history of breast cancer diagnosed at age 28, which is under 50 years, and triple-negative breast cancer status. Additionally, the presence of breast cancer in the patient's mother at age 48 contributes to the criteria. Therefore, the patient is eligible for testing for high-penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Clinical History: Case number 16 of invasive breast carcinoma.

Age of Diagnosis: 28

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncles
- Generation II: Mother diagnosed with breast cancer at age 48.
- Generation III: The patient, diagnosed with breast cancer at age 28.
- Generation IV: Children, Nieces, Nephews

Notes:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at age 28., Triple-negative breast cancer: The patient has ER-negative, PR-negative, and HER2-negative status., Family history of breast cancer: The patient's mother was diagnosed with breast cancer at age 48.

Conclusion

The patient meets multiple NCCN testing criteria, including early-onset breast cancer, triple-negative breast cancer, and a family history of breast cancer. Given this information, the patient is eligible for genetic testing for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2. It is recommended to proceed with the testing to assess hereditary cancer risk and inform management strategies.

Medical Report

Pathology Report Summary

Clinical History: Patient diagnosed with breast cancer, age 45.

Diagnosis: Invasive ductal carcinoma, grade 3, and ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER-2/neu: Positive

Pedigree Analysis

Generations:

- Generation I: No specific details noted.
- Generation II: Notation of Ashkenazi Jewish ancestry.
- Generation III: The patient diagnosed with ER+, PR+, HER2+ breast cancer at age 45., Family history indicates Ashkenazi Jewish ancestry.
- Generation IV: Children

Image Description:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: The patient meets this criterion as they were diagnosed at age 45., Ancestry: Ashkenazi Jewish ancestry: The patient meets this criterion due to noted Ashkenazi Jewish ancestry.

Conclusion

The patient qualifies for NCCN testing criteria for high-penetrance breast cancer susceptibility genes due to the diagnosis of breast cancer at age 45 and Ashkenazi Jewish ancestry. Testing for BRCA1, BRCA2, and other relevant genes is recommended based on the provided criteria.

Medical Report

Pathology Report Summary

Case Number: 17

Age of Diagnosis: 45

Diagnosis: Invasive ductal carcinoma, grade 3, high-grade invasive ductal carcinoma with ductal carcinoma in situ (DCIS) and comedo necrosis.

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Positive
- HER2 Status: Positive

Family History: Ashkenazi Jewish ancestry

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Aunts, Great Uncle
- Generation II: The patient's mother - Unknown cancer history, The patient's father - Unknown cancer history
- Generation III: The patient, diagnosed with breast cancer at age 45., Ashkenazi Jewish ancestry noted.
- Generation IV: Children - No detailed information, Nieces, Nephews - No detailed information

Image:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer before age 50., Ancestry: Ashkenazi Jewish ancestry.

Conclusion

The patient is eligible for NCCN testing criteria based on both their personal history of breast cancer, with diagnosis before age 50, and their Ashkenazi Jewish ancestry. Further genetic testing for BRCA1, BRCA2, and other susceptibility genes is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of high-grade invasive ductal carcinoma, for review.

Case Number: 18

Age of Diagnosis: 75

Diagnosis: Invasive ductal carcinoma, grade 3 with high-grade DCIS and comedo necrosis.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Affected patient with breast cancer at age 75.
- Generation IV: Children, Nieces, Nephews

Family History:

NCCN Testing Criteria Assessment

Criteria: Any age with Triple-negative breast cancer: Not met, as the patient has HER2 positive status., Any age with ≥ 1 close blood relative with breast cancer at age ≤ 50 : Not met, as there is no family history documented., Other family history criteria: Not met, as no relevant family history is noted.

Conclusion

The patient does not meet the NCCN testing criteria for high-penetrance breast cancer susceptibility genes based on the current information. The lack of triple-negative status and absence of relevant family history do not support eligibility for genetic testing under the provided guidelines.

Medical Report

Pathology Report Summary

Clinical History: A 75-year-old individual diagnosed with breast cancer.

Specimen: Not specified.

Diagnosis: Invasive ductal carcinoma, grade 3 with high-grade DCIS and comedo necrosis.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I: No notable occurrences.
- Generation II: No noted family history of cancer.
- Generation III: The patient, diagnosed with breast cancer at 75.
- Generation IV: No offspring information provided.

Visual Representation:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Not met, diagnosed at 75., Triple-negative breast cancer: Not met, HER2 is positive., Family history of breast cancer: Not met, as there is no family history.

Conclusion

The patient does not meet the NCCN criteria for genetic testing based on the current guidelines, as there is no family history and the cancer was diagnosed above the age of 50. The absence of triple-negative receptor status also excludes this criterion.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Breast tissue biopsy.

Diagnosis: Invasive ductal carcinoma, grade 3 with Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Grandfather: Metastatic Prostate Cancer
- Generation II: Father
- Generation III: Patient: Diagnosed with invasive ductal carcinoma at age 50, ER-positive, PR-negative, HER2-negative.

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at age 50, meeting this criterion., Family history of metastatic prostate cancer: The presence of metastatic prostate cancer in the patient's paternal grandfather supports consideration for testing.

Conclusion

The patient meets the NCCN criteria for genetic testing consideration, given the diagnosis of breast cancer at age 50 and a family history of metastatic prostate cancer. Testing for high-penetrance breast cancer susceptibility genes, including BRCA1, BRCA2, and others, is recommended. The familial link to prostate cancer further supports this assessment.

Medical Report

Pathology Report Summary

Case Number: 19

Age of Diagnosis: 50

Diagnosis:

- Invasive Ductal Carcinoma: Grade 3, High-grade
- Ductal Carcinoma In Situ: High-grade with comedo necrosis

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Negative
- HER2: Negative

Pedigree Analysis

Generations:

- Generation I: Grandfather: Prostate cancer, metastatic
- Generation II: Father: Alive/Unknown conditions, Mother: Alive/Unknown conditions
- Generation III: Patient: Breast cancer diagnosed at age 50

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 , ≥ 1 close blood relative with metastatic prostate cancer (paternal grandfather)

Conclusion

The patient meets the NCCN testing criteria for high-penetrance breast cancer susceptibility genes due to a personal history of breast cancer diagnosed at age 50 and a family history of metastatic prostate cancer in a close blood relative (paternal grandfather). Genetic testing for BRCA1, BRCA2, and other applicable genes is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer

Specimen: Breast tissue biopsy.

Diagnosis: Invasive ductal carcinoma, grade 3 with Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

- The patient's pedigree indicates no family history of breast, ovarian, pancreatic, or prostate cancer.
- There are no reported relatives with early - onset breast cancer, no male breast cancer, and no indications of other cancer syndromes

NCCN Testing Criteria Assessment

Analysis for this patient

- Age at diagnosis (67 years) is above the ≤ 50 threshold.
- Tumor is HER2 - positive (thus not triple - negative).
- No indication of multiple primary breast cancers.
- No family history of breast/ovarian/pancreatic/prostate cancer.
- No reported Ashkenazi Jewish ancestry.

Conclusion

Based on the available information and NCCN Guidelines, the patient does not fulfill the criteria for genetic testing for high - penetrance breast cancer susceptibility genes. Therefore, she is not currently eligible under these guidelines.

Medical Report

Pathology Report Summary

Clinical History: Case of breast cancer for review.

Case Number: 20

Age of Diagnosis: 67

Diagnosis: Invasive ductal carcinoma, grade 3 and high-grade DCIS with comedo necrosis.

Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncles
- Generation II: Aunts, Uncles
- Generation III: Siblings, Half-siblings, First cousins
- Generation IV: Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria: Triple-negative breast cancer at any age: Patient tested negative for both ER and PR but was positive for HER2, so this criterion is not met., No family history of breast, ovarian, pancreatic, or prostate cancer, hence other family criteria not applicable.

Conclusion

The patient does not meet the NCCN criteria for high-penetrance breast cancer susceptibility gene testing based on the information provided.

The patient is 67 with HER2-positive breast cancer and no family history of breast or related cancers, which doesn't fulfill NCCN testing criteria.

Medical Report

Pathology Report Summary

Case Number: 21

Age of Diagnosis: 33

Diagnosis: Invasive ductal carcinoma, grade 3; Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptor (ER): Positive
- Progesterone Receptor (PR): Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Mother diagnosed with breast cancer at age ≤ 50 , Maternal aunt diagnosed with breast cancer at age ≤ 50
- Generation III: Patient diagnosed with breast cancer at age 33

Visual Representation:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years: Patient diagnosed at 33.,
Family history of breast cancer: Mother and maternal aunt diagnosed at age ≤ 50 .

Conclusion

The patient is eligible for NCCN testing based on meeting multiple criteria, including personal history with early-onset breast cancer and significant family history. Testing for high-penetrance breast cancer susceptibility genes is advised.

Medical Report

Pathology Report Summary

Case Number: 21

Age of Diagnosis: 33

Diagnosis:

- Invasive Ductal Carcinoma: Grade 3, high-grade invasive ductal carcinoma
- Ductal Carcinoma In Situ (DCIS): High grade with comedo necrosis

Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: The patient's mother diagnosed with breast cancer at age ≤ 50 , The patient's maternal aunt diagnosed with breast cancer at age ≤ 50
- Generation III: The patient, female, diagnosed with invasive ductal carcinoma at age 33

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years, Family history: Mother diagnosed with breast cancer ≤ 50 , Family history: Maternal aunt diagnosed with breast cancer ≤ 50

Conclusion

The patient is eligible for NCCN testing criteria.

The patient meets multiple criteria: diagnosed with breast cancer at age 33, and has a family history of breast cancer with both mother and maternal aunt diagnosed at or before age 50. Recommend genetic testing for high-penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Case Number: 22

Age of Diagnosis: 29

Family History: paternal cousin ovarian cancer

Diagnosis:

- Invasive Ductal Carcinoma: Grade 3
- Ductal Carcinoma In Situ (DCIS): High grade with comedo necrosis

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Positive

Pedigree Analysis

Description:

Image Description:

Generations:

- Generation II: The patient's paternal cousin with ovarian cancer diagnosed at age 40.
- Generation III: The patient, a female, diagnosed with invasive ductal carcinoma at age 29. The cancer is noted as ER-positive, PR-positive, and HER2-positive.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at 29., Family history of ovarian cancer: Paternal cousin diagnosed with ovarian cancer.

Conclusion

The patient is eligible for NCCN testing criteria due to the personal history of breast cancer diagnosed at 29 and a family history of ovarian cancer in a paternal cousin. Therefore, genetic testing for high-penetrance breast cancer susceptibility genes is recommended.

Medical Report

Pathology Report Summary

Case Number: 22

Age of Diagnosis: 29

Family History: Paternal cousin with ovarian cancer

Diagnosis:

- Invasive Ductal Carcinoma: Grade 3, High-grade
- Ductal Carcinoma In Situ (DCIS): High grade with comedo necrosis

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Aunts, Great Uncle
- Generation II: Paternal cousin with ovarian cancer diagnosed at age 40
- Generation III: The patient diagnosed with high-grade invasive ductal carcinoma at age 29
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at 29., Family history: ≥ 1 close blood relative with ovarian cancer. The patient's paternal cousin had ovarian cancer.

Conclusion

The patient meets the NCCN testing criteria due to being diagnosed with breast cancer under the age of 50 and having a close relative with ovarian cancer. These factors highlight the potential for hereditary breast and ovarian cancer syndrome, suggesting genetic testing for high-penetrance breast cancer susceptibility genes is warranted.

Medical Report

Pathology Report Summary

Case Number: 23

Patient Age at Diagnosis: 48 years

Diagnosis:

Invasive ductal carcinoma, Grade 3 (high - grade invasive ductal carcinoma)

Ductal carcinoma in situ (DCIS), high grade with comedo necrosis

Hormone Receptor Status:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Negative

(Overall: Triple - Negative Breast Cancer)

Pedigree Analysis

Brother with male breast cancer diagnosed at age 53.

No other significant family history of breast/ovarian/pancreatic/prostate cancer is noted.

NCCN Testing Criteria Assessment

Criteria: Patient's age at diagnosis is 48 → Meets this criterion.

The patient's tumor is ER-, PR-, HER2- → Meets this criterion.

Brother had male breast cancer → Meets this criterion.

Conclusion

The patient is eligible for genetic testing based on her personal history of triple - negative breast cancer diagnosed at age 48 and having a brother with male breast cancer.

Medical Report

Pathology Report Summary

Case Number: 23

Age at Diagnosis: 48 years

Key Pathological Findings:

High - grade invasive ductal carcinoma (Grade 3)

High - grade DCIS (ductal carcinoma in situ) with comedo necrosis

Hormone Receptor Evaluation:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Negative

Overall Classification: Triple - Negative Breast Cancer

Pedigree Analysis

Brother diagnosed with male breast cancer at age 53

No other significant family history of breast, ovarian, pancreatic, or prostate cancer

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 years: Met, as the patient was diagnosed at age 32., Triple-negative breast cancer: Not met, as the patient is ER-positive and PR-positive.

Conclusion

The patient is eligible for NCCN testing criteria due to the diagnosis of breast cancer at age 32, well below the age of 50.

Medical Report

Pathology Report Summary

Clinical History: Case number 24 - diagnosed with breast cancer at age 55.

Diagnosis: Invasive ductal carcinoma, grade 3 with high-grade ductal carcinoma in situ (DCIS) and comedo necrosis.

Tumor Characteristics:

- Estrogen Receptors: Positive
- Progesterone Receptors: Negative
- HER2: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Father diagnosed with pancreatic cancer at age 65.
- Generation III: Patient diagnosed with breast cancer at age 55, ER-positive, PR-negative, HER2-negative.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 y: Not met (diagnosed at 55)., Any age with Triple-negative breast cancer: Not met., Any age with ≥ 1 close blood relative with pancreatic cancer: Met (father diagnosed with pancreatic cancer at age 65).

Conclusion

The patient meets the NCCN testing criteria for having a close blood relative with pancreatic cancer. This warrants consideration for genetic testing for high-penetrance breast cancer susceptibility genes. Given the combination of personal diagnosis and family history, genetic counseling and further testing, such as for BRCA1 and BRCA2, may be beneficial.

Medical Report

Pathology Report Summary

Patient Identifier: Case 24

Age of Diagnosis: 55

Diagnosis:

- Invasive Ductal Carcinoma: {'Grade': 3, 'Description': 'High-grade invasive ductal carcinoma'}
- Ductal Carcinoma In Situ (DCIS): {'Grade': 'High', 'Features': 'Comedo necrosis'}

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Family History:

Generation I:

Generation II:

- Father: Pancreatic cancer at age 65

Generation III:

- Patient: Breast cancer diagnosed at age 55. ER-positive, PR-negative, HER2-negative.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age > 50 y, Any age with ≥ 1 close blood relative with pancreatic cancer

Conclusion

The patient qualifies for NCCN testing criteria.

Based on the patient's diagnosis of breast cancer at age 55 and a family history of pancreatic cancer in a close blood relative (father), the patient is eligible for genetic testing according to the NCCN Guidelines. Testing for high-penetrance breast cancer susceptibility genes is warranted.

Medical Report

Pathology Report Summary

Case Number: 25

Age of Diagnosis: 75

Diagnosis: Invasive carcinoma with lobular features (Grade 3)

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Positive
- HER2 Status: Positive

Family History: No family history

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: The affected patient, female, diagnosed at 75., No siblings with cancer noted.
- Generation IV: Children, Nieces, Nephews

Image Description:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Not met, diagnosed at 75., Triple-negative breast cancer: Not met, PR positive and HER2 positive., Multiple primary breast cancers: Not applicable., Lobular breast cancer with family history of diffuse gastric cancer: Not applicable., Male breast cancer: Not applicable., Ashkenazi Jewish ancestry: Not reported., Close blood relative with early-onset breast cancer, male breast cancer, ovarian, pancreatic or metastatic prostate cancer: Not applicable., ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family: Not reported.

Conclusion

The patient does not meet the NCCN criteria for genetic testing for high-penetrance breast cancer susceptibility genes. The lack of significant family history and a diagnosis at age 75 reduce the probability of a hereditary cancer syndrome. Thus, genetic testing according to NCCN Guidelines is not warranted in this case. Continued monitoring and routine care are advised.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Diagnosis: Invasive carcinoma with lobular features (Grade 3).

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Positive
- HER2: Positive

Note: The HER2-positive status may suggest more aggressive treatment options including targeted therapy.

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: Patient (Case 25), female, diagnosed with breast cancer at age 75.
- Generation IV: Children, Nieces, Nephews

Textual Analysis:

NCCN Testing Criteria Assessment

Criteria Met: None

Analysis: The patient, a 75-year-old female, has a family history of breast cancer. Her mother (Generation II) was diagnosed with breast cancer at age 65. The patient (Generation III) was diagnosed with breast cancer at age 75. There is no history of breast cancer in the patient's father (Generation II) or her siblings (Generation II). The patient's mother (Generation II) has one child (the patient, Generation III) and one niece (Generation III). The patient's mother (Generation II) has one child (the patient, Generation III) and one niece (Generation III). The patient's mother (Generation II) has one child (the patient, Generation III) and one niece (Generation III).

Conclusion

The patient is not eligible for NCCN testing under the current criteria. She does not meet age-related or familial-based conditions listed for genetic testing of high-penetrance breast cancer

susceptibility genes. No action regarding genetic testing is recommended based on the provided information.

Medical Report

Pathology Report Summary

Clinical History: Invasive carcinoma with lobular features, diagnosed at age 64.

Specimen: Not specified.

Diagnosis: Invasive carcinoma with lobular features (Grade 2).

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Paternal Aunt with ovarian cancer diagnosed at age 60.
- Generation III: Case 26, female, diagnosed with breast cancer at age 64.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria: Any age with ≥ 1 close blood relative with ovarian cancer: The patient has a paternal aunt diagnosed with ovarian cancer.

Conclusion

The patient is eligible for NCCN testing criteria due to having a close blood relative with ovarian cancer. This fulfills one of the NCCN testing criteria for high-penetrance breast cancer susceptibility genes, suggesting further genetic testing could be justified.

Medical Report

Pathology Report Summary

Case Number: 26

Age at Diagnosis: 64 years

Pathological Findings: Invasive breast carcinoma with lobular features (Grade 2)

Hormone Receptor Status:

- ER (Estrogen Receptor): Positive
- PR (Progesterone Receptor): Positive
- HER2 (Human Epidermal Growth Factor Receptor 2): Negative

Pedigree Analysis

Paternal aunt affected by ovarian cancer

No other significant family history of breast, pancreatic, prostate, or other hereditary cancers is noted

NCCN Testing Criteria Assessment

Criteria: Any close relative with ovarian cancer

Because the patient has a paternal aunt with ovarian cancer, she meets the relevant NCCN criterion that supports testing for high - penetrance breast cancer genes (e.g., BRCA1, BRCA2).

Conclusion

Based on her family history of a paternal aunt with ovarian cancer, the patient meets NCCN criteria and is therefore eligible for genetic testing.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer for review.

Specimen: Not specified.

Diagnosis: Invasive carcinoma with lobular features, Grade 2.

Tumor Characteristics:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Paternal Grandfather with prostate cancer.
- Generation II: Two uncles from the paternal side with prostate cancer.
- Generation III: Case 27, diagnosed with breast cancer at age 38, ER and PR positive, HER2 negative.

Description:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years: Patient diagnosed at 38., Family history of prostate cancer: Paternal grandfather and two uncles affected.

Conclusion

The patient is eligible for NCCN testing based on personal history of breast cancer diagnosis at age 38 and the presence of 3 family members with prostate cancer. Testing for high-penetrance breast cancer susceptibility genes (e.g., BRCA1, BRCA2) is recommended. Further evaluation of the prostate cancer aggressiveness in the family would enhance risk assessment.

Medical Report

Pathology Report Summary

Case Number: 27

Age of Diagnosis: 38

Diagnosis: Invasive carcinoma with lobular features (Grade 2).

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Paternal Grandfather with prostate cancer
- Generation II: Two uncles with prostate cancer
- Generation III: Patient diagnosed with breast cancer at age 38 (ER-positive, PR-positive, HER2-negative)

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 y. Three paternal relatives (grandfather, two uncles) with prostate cancer → Meets Criterion.

Conclusion

Because this patient was diagnosed with breast cancer before age 50 and has a paternal family history of three prostate cancers, she satisfies multiple NCCN testing criteria. She is therefore eligible for genetic testing for high - penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Clinical History: Case 28 of breast cancer.

Specimen: Not specified in provided information.

Diagnosis: Invasive carcinoma with lobular features, Grade 1.

Receptor Status:

- Estrogen Receptors (ER): Negative
- Progesterone Receptors (PR): Negative
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: The patient, female, diagnosed at age 45 with triple-negative lobular breast cancer.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 years: Met, as the patient was diagnosed at age 45., Triple-negative breast cancer: Met, as the patient is ER-, PR-, HER2-.

Conclusion

The patient meets multiple NCCN criteria for genetic testing. Diagnosed with breast cancer at age 45 and having triple-negative breast cancer justifies testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2, CDH1, PALB2, PTEN, STK11, and TP53. Given the absence of a significant family history, the personal diagnosis is crucial for eligibility.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Right breast, tru cut biopsy.

Diagnosis: Invasive carcinoma with lobular features (Grade 1).

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Uncle
- Generation III: The patient, diagnosed with breast cancer at age 45, with no familial history of breast cancer noted in the image.
- Generation IV: Children, Nieces, Nephews

Notes:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 : The patient was diagnosed at age 45, which meets this criterion., Triple-negative breast cancer: The patient has triple-negative status (ER-, PR-, HER2-), which qualifies for testing., Family history is negative for other related cancers or genetic risk noted in the pedigree, so no additional criteria met based on family history.

Conclusion

The patient is eligible for NCCN Criteria due to a personal history of breast cancer diagnosed at age 45 and triple-negative breast cancer status. Despite the absence of a notable family history of related cancers, the triple-negative status and age at diagnosis alone justify genetic testing consideration.

Medical Report

Pathology Report Summary

Case Number: 29

Age of Diagnosis: 55

Family History: Sister diagnosed with breast cancer at age 50

Diagnosis: Invasive carcinoma with lobular features (Grade 1)

Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: The patient, diagnosed with breast cancer at age 55, with ER-positive, PR-negative, HER2-positive status
- Generation IV: Sister diagnosed with breast cancer at age 50

Description:

NCCN Testing Criteria Assessment

Criteria Met: Age of diagnosis > 50 years, does not meet the personal history criterion for testing., Family history fulfills NCCN criteria due to the sister being diagnosed with breast cancer at age ≤50.

Conclusion

The patient is eligible for NCCN testing based on the family history criterion, as the sister was diagnosed with breast cancer at age 50.

Given the patient's family history, they meet the testing criteria for genes related to high-penetrance breast cancer susceptibility, despite their diagnosis occurring after the age cutoff for personal history. Recommended testing for BRCA1, BRCA2, and potentially other genes is warranted.

Medical Report

Pathology Report Summary

Case Number: 29

Age of Diagnosis: 55

Family History: Sister diagnosed with breast cancer at age 50

Diagnosis: Invasive carcinoma with lobular features (Grade 1)

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: The patient, diagnosed with breast cancer at age 55. ER-positive, PR-negative, HER2-positive., The patient's sister, diagnosed with breast cancer at age 50.
- Generation IV: Children, Nieces, Nephews

Visual Representation:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 : Not met, patient diagnosed at 55.,

Family history: ≥ 1 close blood relative with breast cancer at age ≤ 50 : Met, patient's sister diagnosed at 50.

Conclusion

The patient is eligible for NCCN testing criteria based on having a sister diagnosed with breast cancer at age 50. This family history aligns with the testing criteria for high-penetrance breast cancer susceptibility genes. Thus, genetic testing for BRCA1, BRCA2, and other related genes is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Not specified

Diagnosis: Invasive carcinoma with lobular features (Grade 3).

Hormone Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER-2/neu: Negative

Pedigree Analysis

Visual Representation:

Generations:

- Generation I: GPs, Aunts, Great Uncle
- Generation II: Parents, Aunts
- Generation III: Patient: diagnosed with breast cancer at age 67, ER/PR/HER2 negative.
- Generation IV: Children, Nieces, Nephews

Analysis:

NCCN Testing Criteria Assessment

Criteria Met: Triple-negative breast cancer at any age., Ashkenazi Jewish ancestry.

Conclusion

The patient is eligible for NCCN testing criteria based on having triple-negative breast cancer and Ashkenazi Jewish ancestry. This qualifies the patient for evaluation for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2. Genetic counseling is recommended prior to testing to discuss potential outcomes and implications.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, age of diagnosis is 67.

Diagnosis: Invasive carcinoma with lobular features (Grade 3).

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Aunts, Great Uncles
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient with breast cancer, diagnosis at age 67, Ashkenazi Jewish ancestry, triple-negative breast cancer.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer., Any age with Ancestry: Ashkenazi Jewish ancestry.

Conclusion

The patient is eligible for NCCN criteria testing.

The patient meets eligibility based on having triple-negative breast cancer and being of Ashkenazi Jewish ancestry. Genetic testing for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2 is indicated.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer for review.

Specimen: Not explicitly stated, but inferred as breast tissue.

Diagnosis: Invasive carcinoma with lobular features (Grade 2).

Age of Diagnosis: 65

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Positive
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Aunts, Uncles
- Generation III: Patient (Case 31): Diagnosed with breast cancer at age 65, ER/PR/HER2 positive.
- Generation IV: Children, Nieces, Nephews

Visual Representation:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Not met (diagnosed at 65)., Triple-negative breast cancer: Not met (ER, PR, HER2 positive)., Multiple primary breast cancers: Not met., Lobular breast cancer with personal or family history of diffuse gastric cancer: Not met., Male breast cancer: Not applicable., Ancestry: Ashkenazi Jewish ancestry: Not mentioned., ≥ 1 close blood relative with breast cancer at age ≤ 50 : Not applicable., ≥ 1 close blood relative with male breast cancer: Not applicable., ≥ 1 close blood relative with ovarian cancer: Not applicable., ≥ 1 close blood relative with pancreatic cancer: Not applicable., ≥ 1 close blood relative with prostate cancer with metastatic, or high- or very-high-risk group: Not mentioned., ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family: Not met.

Conclusion

The patient does not meet the NCCN criteria for genetic testing. The diagnosis at age 65, along with ER/PR/HER2 positivity, and the absence of significant family history of breast or other related cancers, suggest that genetic testing based on NCCN Guidelines is not warranted at this time.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Not specified in the document.

Diagnosis: Invasive carcinoma with lobular features (Grade 2).

Hormone Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Positive
- HER2 Status: Positive

Note: No additional treatment suggestions provided in the document.

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
 - Generation II: Aunts, Uncles
 - Generation III: The patient, with a diagnosis of breast cancer at age 65., No noted family history of breast cancer or related conditions.
 - Generation IV: Children, Nieces, Nephews
- NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Not met, as the diagnosis was made at age 65., Triple-negative breast cancer: Not met, as hormone receptor status is positive., Multiple primary breast cancers: Not indicated., Lobular breast cancer with personal or family history of diffuse gastric cancer: Not indicated., Male breast cancer: Not applicable., Ancestry: Ashkenazi Jewish ancestry: Not indicated., ≥ 1 close blood relative with breast cancer at age ≤ 50 : Not indicated., ≥ 1 close blood relative with male breast cancer: Not indicated., ≥ 1 close blood relative with ovarian cancer: Not indicated., ≥ 1 close blood relative with pancreatic cancer: Not indicated., ≥ 1 close blood relative with prostate cancer with metastatic or high-risk group: Not indicated., ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family: Not indicated.

Conclusion

The patient does not meet the NCCN criteria for genetic testing based on the provided clinical information and family history. Diagnosis was at age 65 with no significant family history meeting NCCN Guidelines. The positive hormone receptor status further excludes her from some criteria. Genetic testing might not be warranted strictly based on NCCN criteria, but decisions might consider other clinical factors.

Medical Report

Pathology Report Summary

Case Number: 32

Patient Age at Diagnosis: 34 years

Diagnosis: Invasive breast carcinoma with lobular features (Grade 2)

Hormone Receptor Status:

ER (Estrogen Receptor): Negative

PR (Progesterone Receptor): Negative

HER2 (Human Epidermal Growth Factor Receptor 2): Negative

(Overall: Triple - Negative Breast Cancer)

Pedigree Analysis

Father was diagnosed with pancreatic cancer.

No other significant family history (e.g., breast, ovarian, prostate) reported.

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer diagnosed at age ≤ 50 , Patient is 34 at diagnosis →

Meets this criterion. Triple - negative breast cancer at any age, Tumor is ER-, PR-, HER2- →

Meets this criterion. Father diagnosed pancreatic cancer Meets criteria.

Conclusion

The patient meets the NCCN testing criteria for high-penetrance breast cancer susceptibility genes based on personal history of early-onset breast cancer, triple-negative status, and family history of pancreatic cancer. Genetic testing for BRCA1, BRCA2, and other relevant genes is warranted.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer for review, with a family history of pancreatic cancer.

Specimen: Details not specified in document.

Diagnosis: Invasive carcinoma with lobular features (Grade 2).

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER-2/neu: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Father: pancreatic cancer.
- Generation III: Patient: diagnosed with invasive carcinoma with lobular features at age 34.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: Yes, diagnosed at 34., Triple-negative breast cancer: Yes, ER, PR, and HER2 negative., Family history of pancreatic cancer: Yes, father had pancreatic cancer.

Conclusion

The patient meets the NCCN testing criteria for high-penetrance breast cancer susceptibility genes based on personal history of early-onset breast cancer, triple-negative status, and family history of pancreatic cancer. Genetic testing for BRCA1, BRCA2, and other relevant genes is warranted.

Medical Report

Pathology Report Summary

Clinical History: Case 33, diagnosis at age 29.

Specimen: Not specified.

Diagnosis: Invasive carcinoma with lobular features (Grade 1). Multiple primary breast cancer.

Hormone Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Negative
- HER2: Negative

Note: No family history mentioned.

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: The patient, a female, diagnosed with invasive carcinoma at age 29.
- Generation IV: Children, Nieces, Nephews

Description:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at 29, meeting this criterion., Multiple primary breast cancers: Met because the diagnosis includes multiple primary breast cancers., Triple-negative breast cancer: Not applicable., Lobular breast cancer with personal or family history of diffuse gastric cancer: No family history noted., Male breast cancer: Not applicable., Ashkenazi Jewish ancestry: Not applicable., Close blood relative with breast, ovarian, pancreatic, or prostate cancer: No noted family history.

Conclusion

The patient meets the NCCN testing criteria due to her diagnosis of breast cancer at age 29 and multiple primary breast cancers, despite no noted family history of related cancers. Genetic testing for high-penetrance breast cancer susceptibility genes is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of invasive carcinoma with lobular features.

Specimen: Breast cancer case report.

Diagnosis: Invasive carcinoma with lobular features (Grade 1), Multiple primary breast cancer.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Negative
- HER-2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: The patient diagnosed with breast cancer at age 29 with no family history.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at 29, meeting this criterion., Multiple primary breast cancers: The patient has multiple primary breast cancers, meeting this criterion.

Conclusion

The patient meets the NCCN criteria for genetic testing due to their young age at diagnosis (29) and having multiple primary breast cancers. Despite no family history of breast cancer, these factors warrant consideration for genetic testing for high-penetrance breast cancer susceptibility genes like BRCA1 and BRCA2.

Medical Report

Pathology Report Summary

Case Number: 34

Age of Diagnosis: 56

Family History: Sister with ovarian cancer

Diagnosis: Invasive breast carcinoma of no special type, grade 3. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient with breast cancer diagnosed at age 56, ER-negative, PR-negative, HER2-negative., Sister with ovarian cancer diagnosed at age 55.
- Generation IV: Children, Nieces, Nephews

Visual Link:

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer., Any age with ≥ 1 close blood relative with ovarian cancer.

Conclusion

The patient is eligible for NCCN testing criteria. The presence of triple-negative breast cancer and a close blood relative with ovarian cancer justify consideration for genetic testing for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2. It is recommended to consult with a genetic counselor for further assessment.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Diagnosis: Invasive breast carcinoma of no special type, grade 3; Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Aunts, Uncles
- Generation II: Patient's sister diagnosed with ovarian cancer at age 55.
- Generation III: The patient, diagnosed with breast cancer at age 56.
- Generation IV: Children, Nieces, Nephews

Visual Representation:

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer., Any age with ≥ 1 close blood relative with ovarian cancer.

Criteria Not Met: Personal history of breast cancer with age ≤ 50 y.

Conclusion

The patient is eligible for the NCCN testing criteria due to the diagnosis of triple-negative breast cancer and having a sister with ovarian cancer. This suggests testing for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2.

Medical Report

Pathology Report Summary

Clinical History: Case of invasive breast carcinoma.

Specimen: Not specified in the document.

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle with male breast cancer
- Generation II: Paternal uncle with male breast cancer
- Generation III: Affected patient, diagnosed at age 68 with invasive breast carcinoma.
- Generation IV: Children, Nieces, Nephews

Image Link or Description:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with diagnosis at age >50., Family history: Paternal uncle with male breast cancer.

Conclusion

The patient is eligible for NCCN genetic testing criteria due to the family history of male breast cancer in a paternal uncle, despite the personal cancer diagnosis occurring at an age greater than 50. Genetic testing for high-penetrance breast cancer susceptibility genes, such as BRCA1 and BRCA2, is recommended given the family history and positive hormone receptor status.

Medical Report

Pathology Report Summary

Case Number: 35

Age of Diagnosis: 68

Family History: Paternal Uncle with male breast cancer

Diagnosis: Invasive breast carcinoma of no special type, grade 1.

Ductal Carcinoma In Situ (DCIS): Intermediate grade with a solid pattern.

Hormone Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Positive
- HER2 Status: Positive

Pedigree Analysis

Description:

Generations:

- Generation I: Paternal Grandparents
- Generation II: Paternal Uncle diagnosed with male breast cancer
- Generation III: The patient diagnosed at age 68 with ER+, PR+, HER2+ breast cancer

NCCN Testing Criteria Assessment

Criteria Met: Any age with a close blood relative with male breast cancer.

Conclusion

The patient is eligible for NCCN genetic testing criteria due to a family history of male breast cancer in a close blood relative (paternal uncle). Testing for high-penetrance breast cancer susceptibility genes such as BRCA1 and BRCA2 is recommended.

Medical Report

Pathology Report Summary

Clinical History: Case of breast cancer, for review.

Case Number: 36

Age of Diagnosis: 50

Diagnosis: Invasive breast carcinoma of no special type, grade 2.

Additional Findings: Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Father with metastatic prostate cancer., Paternal uncle with metastatic prostate cancer.
- Generation III: Female diagnosed with breast cancer at age 50 (case 36).
- Generation IV:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at age 50, meeting this criterion., Triple-negative breast cancer: The patient is ER-negative and PR-negative, but HER2-positive (not triple-negative)., ≥ 1 close blood relative with prostate cancer with metastatic disease: Both father and paternal uncle have metastatic prostate cancer, meeting this criterion.

Conclusion

The patient is eligible for NCCN testing criteria.

Given the patient's age at diagnosis (50) and family history of metastatic prostate cancer in close relatives, the criteria for testing for high-penetrance breast cancer susceptibility genes are met. Testing for genes such as BRCA1, BRCA2, and others is recommended.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Biopsy analysis.

Diagnosis: Invasive breast carcinoma of no special type, grade 2.

DCIS: Intermediate grade ductal carcinoma in situ with a solid pattern.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I: Grandparents, some unknown health details.
- Generation II: Father and paternal uncle with metastatic prostate cancer.
- Generation III: Patient diagnosed with breast cancer at age 50, ER-negative, PR-negative, HER2-positive.
- Generation IV: Information about offspring or other close relatives not detailed.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years., HER2-positive status alone does not directly factor, but other risk factors align.

Family Criteria: Paternal family history includes metastatic prostate cancer, but no females diagnosed with breast cancer at ≤ 50 years.

Conclusion

The patient meets the NCCN testing criteria based on a personal history of breast cancer at age 50, and HER2-positive status. Family history further supports genetic testing consideration due to metastatic prostate cancer in close relatives. Given these factors, genetic testing for high-penetrance genes such as BRCA1 and BRCA2 is warranted.

Medical Report

Pathology Report Summary

Case Number: 37

Age of Diagnosis: 45

Family History: Maternal cousin breast cancer at age 49

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Maternal cousin with breast cancer diagnosed at age 49
- Generation III: The patient, a female, diagnosed with invasive breast carcinoma at age 45, Triple-negative (ER-, PR-, HER2-)

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years (diagnosed at 45), Any age with Triple-negative breast cancer, Any age with ≥ 1 close blood relative with breast cancer at age ≤ 50 (maternal cousin diagnosed at 49)

Conclusion

The patient is eligible for NCCN testing based on multiple criteria met: personal history of breast cancer diagnosed before the age of 50, triple-negative breast cancer, and a maternal cousin with breast cancer diagnosed before age 50. Comprehensive genetic testing for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2, is recommended.

Medical Report

Pathology Report Summary

Case Number: 37

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Age of Diagnosis: 45

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Aunts, Uncles
- Generation III: Patient, diagnosed with breast cancer at age 45 with triple-negative status., Maternal cousin with breast cancer at age 49.
- Generation IV: Children, Nieces, Nephews

Significant Familial Links:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years., Any age with Triple-negative breast cancer., Close blood relative (maternal cousin) with breast cancer at age ≤ 50 .

Conclusion

The patient meets multiple NCCN criteria for testing, including being diagnosed with breast cancer at age 45, triple-negative status, and a close blood relative with early-onset breast cancer. Therefore, the patient is eligible for genetic testing following NCCN Guidelines for high-penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Clinical History: Case 38, 65-year-old patient diagnosed with breast cancer.

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient diagnosed with breast cancer at age 65, ER-positive, PR-positive, HER2-negative.
- Generation IV: Children, Nieces, Nephews

Family Health History:

NCCN Testing Criteria Assessment

Criteria Analysis: Personal history of breast cancer with age ≤ 50 y: Not met, patient diagnosed at 65 years., Triple-negative breast cancer: Not applicable, patient is ER-positive and PR-positive., Multiple primary breast cancers: Not indicated for this patient., Significant family history: No additional noted indicators such as male breast cancer or specific ancestries for high-risk consideration.

Conclusion

The patient does not meet the NCCN criteria for genetic testing based on the provided information. No strong family history or other criteria suggest further genetic testing under current guidelines.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer for a 65-year-old patient.

Diagnosis: Invasive breast carcinoma of no special type, grade 1. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: The patient, diagnosed with breast cancer at age 65. ER-positive, PR-positive, HER2-negative.
- Generation IV: Children, Nieces, Nephews

VisualRepresentation:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 : Not met, diagnosed at 65., Triple-negative breast cancer: Not met, as HER2 status is negative but ER and PR are positive., Multiple primary breast cancers: Not met., Lobular breast cancer with a family history of diffuse gastric cancer: Not met., Male breast cancer: Not met., Ashkenazi Jewish ancestry: Not mentioned., Close blood relative with breast cancer at age ≤ 50 : Not noted., Close blood relative with male breast cancer: Not noted., Close blood relative with ovarian cancer: Not noted., Close blood relative with pancreatic cancer: Not noted., Close blood relative with prostate cancer with metastatic, or high- or very-high-risk group: Not noted., ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family: Not noted.

Conclusion

The patient does not meet the NCCN criteria for high-penetrance breast cancer susceptibility testing. Given the absence of a significant family history of cancers outlined in the criteria, genetic testing based on NCCN Guidelines may not be necessary. Clinical judgment should consider other factors and individual circumstances.

Medical Report

Pathology Report Summary

Case Number: 39

Age of Diagnosis: 30

Diagnosis: Invasive breast carcinoma of no special type, grade 2.

Ductal Carcinoma In Situ (DCIS): Intermediate grade with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncles
- Generation II: Mother
- Generation III: Patient, diagnosed with breast cancer at age 30, ER-negative, PR-negative, HER2-negative., Sister diagnosed with ovarian cancer.
- Generation IV: Children, Nieces, Nephews

Family Health History:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years., Triple-negative breast cancer., Close blood relative (sister) with ovarian cancer.

Conclusion

The patient is eligible for NCCN testing based on multiple criteria: diagnosis of breast cancer at age 30, triple-negative breast cancer status, and a close blood relative (sister) with ovarian cancer. Genetic testing for high-penetrance breast cancer susceptibility genes like BRCA1 and BRCA2 is recommended due to these factors.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Case Number: 39

Age of Diagnosis: 30

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Image Description:

Generations:

- Generation II: The patient, a female, diagnosed with breast cancer at age 30., The patient's sister diagnosed with ovarian cancer.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years., Triple-negative breast cancer (ER negative, PR negative, HER2 negative)., Close blood relative with ovarian cancer (sister).

Conclusion

The patient is eligible for NCCN testing criteria.

The patient's early age of diagnosis, triple-negative breast cancer status, and family history of ovarian cancer (sister) fulfill criteria for genetic testing. It is recommended to test for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2, among others.

Medical Report

Pathology Report Summary

Clinical History: A 29-year-old patient diagnosed with breast cancer.

Specimen: Not specified in the document.

Diagnosis: Invasive carcinoma with lobular features (Grade 1).

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Negative
- HER-2/neu: Positive

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncles
- Generation II: Aunts, Uncles
- Generation III: The patient, diagnosed with breast cancer at age 29., Cousin with breast cancer diagnosed at age 48.
- Generation IV: Children, Nieces, Nephews

Analysis:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer at age ≤ 50 : Met, as the patient was diagnosed at age 29., Close blood relative with breast cancer at age ≤ 50 : Met, as the cousin was diagnosed at age 48.

Conclusion

The patient meets the NCCN criteria for genetic testing due to personal history of early-onset breast cancer (diagnosed at age 29) and having a cousin who was diagnosed with breast cancer at age 48. Genetic testing for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2, is recommended to better understand the hereditary risks and inform management strategies.

Medical Report

Pathology Report Summary

Clinical History: Patient diagnosed with invasive carcinoma with lobular features.

Specimen: Not specified.

Diagnosis: Invasive carcinoma with lobular features (Grade 1).

Receptor Status:

- ER Status: Positive
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncles
- Generation II: Aunts, Uncles
- Generation III: Patient diagnosed with breast cancer at age 29, ER-positive, PR-negative, HER2-positive., Cousin with breast cancer diagnosed at age 48.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years: The patient was diagnosed at age 29., Close blood relative with breast cancer at age ≤ 50 : The patient's cousin was diagnosed at age 48.

Conclusion

The patient meets the NCCN testing criteria due to the diagnosis of breast cancer at age 29 and a close relative diagnosed at age 48. Therefore, they are eligible for testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2, etc. This testing can provide important insights into potential hereditary patterns in the family.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Breast biopsy.

Diagnosis: Invasive breast carcinoma of no special type, grade 2, with intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Aunts, Uncles
- Generation III: The affected patient, diagnosed at age 75 with invasive breast carcinoma.
- Generation IV: Children, Nieces, Nephews

Analysis:

NCCN Testing Criteria Assessment

Criteria: Triple-negative breast cancer: The patient is ER-negative and PR-negative but HER2-positive, thus not meeting this criterion., No family history of breast cancer at or below 50 years, male breast cancer, or other related cancers.

Conclusion

The patient is not eligible for the NCCN criteria for high-penetrance breast cancer susceptibility gene testing as none of the listed criteria are met. There is no family history and the cancer type and hormone receptor status do not align with the testing criteria.

Medical Report

Pathology Report Summary

Case Number: 41

Age of Diagnosis: 75

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Aunts, Uncles
- Generation III: Patient: Female, diagnosed with breast cancer at age 75, ER-negative, PR-negative, HER2-positive.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria: Triple-negative breast cancer at any age: The patient is not eligible as the cancer is HER2-positive., Family history: No family history of cancer is mentioned in the document or image.

Conclusion

The patient does not meet the NCCN testing criteria for high-penetrance breast cancer susceptibility genes due to the lack of triple-negative status, no personal history of breast cancer at age ≤ 50 , and no notable family history of related cancers. Given the patient's current age and cancer characteristics, NCCN testing is not indicated.

Medical Report

Pathology Report Summary

Case Number: 42

Age at Diagnosis: 44

Diagnosis: Invasive carcinoma with lobular features (Grade 1)

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Aunt with pancreatic cancer.

No other immediate reports of breast, ovarian, or other related cancers in the family pedigree.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years, Any age with Triple-negative breast cancer, Any age with ≥ 1 close blood relative with pancreatic cancer

Conclusion

The patient meets the NCCN criteria for genetic testing. Specifically, she was diagnosed with breast cancer at age 44, her cancer is triple-negative, and she has a close blood relative (aunt) with pancreatic cancer. It is recommended to consider testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2 among others.

Medical Report

Pathology Report Summary

Case Number: 42

Patient's Age at Diagnosis: 44 years

Diagnosis: Invasive breast carcinoma with lobular features (Grade 1)

Hormone Receptor Status:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Negative

Overall Classification: Triple - Negative Breast Cancer

Pedigree Analysis

Aunt diagnosed with pancreatic cancer

No further significant family history is reported

NCCN Testing Criteria Assessment

Criteria Met:

Diagnosed at age 44 → Meets the ≤ 50 criterion, Breast cancer at or before age 50.

Tumor is triple - negative → Meets criterion, Triple - negative breast cancer at any age.

Aunt with pancreatic cancer → Meets criterion, at least one close blood relative with pancreatic cancer.

Conclusion

Because this patient's breast cancer was diagnosed at age 44, is triple - negative, and she has a close blood relative with pancreatic cancer, she meets multiple NCCN criteria and is therefore eligible for genetic testing.

Medical Report

Pathology Report Summary

Case Number: 43

Age of Diagnosis: 38

Diagnosis:

- Invasive Ductal Carcinoma: High-grade invasive ductal carcinoma, grade 3
- Ductal Carcinoma In Situ (DCIS): High grade with comedo necrosis

Hormone Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Positive

Pedigree Analysis

Visual Representation:

Generations:

- Generation I: Grandfather had metastatic prostate cancer
- Generation II:
- Generation III: Patient diagnosed with ER-positive, PR-positive, HER2-positive breast cancer at age 38
- Generation IV:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years, Family history of metastatic prostate cancer

Conclusion

The patient meets the NCCN testing criteria due to the diagnosis of breast cancer at age ≤ 50 and the family history of metastatic prostate cancer. Testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2, and others is recommended.

Medical Report

Pathology Report Summary

Clinical History: Case of breast cancer, patient aged 38.

Case Number: 43

Diagnosis: Invasive ductal carcinoma, grade 3: High-grade invasive ductal carcinoma; Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors (ER): Positive
- Progesterone Receptors (PR): Positive
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I: Grandfather with metastatic prostate cancer.
- Generation II: Parents: No cancer history indicated.
- Generation III: Patient diagnosed with breast cancer at age 38, ER/PR/HER2 positive.
- Generation IV: No data provided.

Image Description:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer at age ≤ 50 years: The patient was diagnosed at age 38., Family history of prostate cancer with metastatic condition: Grandfather had metastatic prostate cancer.

Conclusion

The patient is eligible for NCCN testing for high-penetrance breast cancer susceptibility genes based on her diagnosis at age 38 and a family history of metastatic prostate cancer in the grandfather. Testing for BRCA1, BRCA2, and other associated genes may be warranted given this history.

Medical Report

Pathology Report Summary

Case Number: 44

Patient Age at Diagnosis: 45 years

Diagnosis: Invasive breast carcinoma with lobular features (Grade 2)

Hormone Receptor Status:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Positive

Pedigree Analysis

Mother diagnosed with breast cancer at age 52.

No other significant family history of breast, ovarian, or related cancers noted.

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer diagnosed at age ≤ 50 , Patient is 45 at diagnosis
→ Meets this criterion. Since the patient's breast cancer was diagnosed at age ≤ 50 , she meets an NCCN testing

Conclusion

The patient is eligible for genetic testing under NCCN Guidelines due to her personal history of breast cancer diagnosed at age ≤ 50 .

Medical Report

Pathology Report Summary

Case Number: 44

Age at Diagnosis: 45 years

Pathological Findings: Invasive breast carcinoma with lobular features (Grade 2)

Hormone Receptor Status:

- ER (Estrogen Receptor): Negative
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Positive
-

Pedigree Analysis

Family History: Mother diagnosed with breast cancer at age 52

No additional pertinent cancers noted in the family

NCCN Testing Criteria Assessment

Patient age of 45 at diagnosis → Meets the ≤ 50 years criterion.

Tumor is HER2 - positive, not triple - negative → Does not meet.

No family history of breast cancer at ≤ 50 (mother was 52) → Does not meet.

Other listed family - history or personal - history criteria are not fulfilled.

Conclusion

Because the patient was diagnosed with breast cancer at age 45 (meeting the NCCN threshold of ≤ 50), she qualifies for genetic testing for high - penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Case Number: 45

Age of Diagnosis: 56

Diagnosis: Invasive ductal carcinoma, grade 3 with high-grade ductal carcinoma in situ (DCIS) and comedo necrosis.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Siblings, Cousins
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria Met: Triple-negative breast cancer at any age.

Criteria Not Met: Personal history of breast cancer with age ≤ 50 y., Multiple primary breast cancers., Lobular breast cancer with personal or family history of diffuse gastric cancer., Male breast cancer., Ancestry: Ashkenazi Jewish ancestry., ≥ 1 close blood relative with breast cancer at age ≤ 50 ., ≥ 1 close blood relative with male breast cancer., ≥ 1 close blood relative with ovarian cancer., ≥ 1 close blood relative with pancreatic cancer., ≥ 1 close blood relative with prostate cancer with metastatic/high/very-high-risk group., ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family including the patient.

Conclusion

The patient is eligible for NCCN testing criteria due to triple-negative breast cancer at any age.

The patient's triple-negative breast cancer status qualifies them for genetic testing under the NCCN Guidelines. No additional family history of breast or related cancers is indicated, aligning with the criteria for triple-negative cases.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer with no family history.

Specimen: Breast tissue, biopsy.

Diagnosis: Invasive ductal carcinoma, grade 3; Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: GPs, Great Aunts, Great Uncle
- Generation II: Parents, Aunts, Uncles
- Generation III: Patient, a female, diagnosed with triple-negative breast cancer at age 56.
- Generation IV: Children, Nieces, Nephews

Family History:

NCCN Testing Criteria Assessment

Criteria: Any age with Triple-negative breast cancer: The patient meets this criterion as she has triple-negative breast cancer.

Conclusion

The patient meets the NCCN testing criteria for genetic testing based on the presence of triple-negative breast cancer, despite having no family history of breast cancer or related conditions. This warrants consideration for testing for high-penetrance breast cancer susceptibility genes such as BRCA1, BRCA2, and others as per NCCN Guidelines.

Medical Report

Pathology Report Summary

Case Number: 46

Patient Age at Diagnosis: 29 years

Diagnosis: Invasive breast carcinoma (Grade 2)

Intermediate - grade Ductal Carcinoma In Situ (DCIS) with a solid pattern

Hormone Receptor Status:

- ER (Estrogen Receptor): Positive
- PR (Progesterone Receptor): Negative
- HER2 (Human Epidermal Growth Factor Receptor 2): Negative

Pedigree Analysis

Sister diagnosed with triple - negative breast cancer at age 45.

No other notable family history of ovarian, pancreatic, or prostate cancer.

NCCN Testing Criteria Assessment

The patient meets two NCCN criteria: personal diagnosis of breast cancer at or before age 50 and having a close blood relative (sister) with breast cancer at age ≤ 50 .

Conclusion

The patient is eligible for genetic testing under NCCN Guidelines based on her personal history of breast cancer at age ≤ 50 and a sister with breast cancer at age 45.

Medical Report

Pathology Report Summary

Case Number: 46

Age of Diagnosis: 29

Diagnosis: Invasive breast carcinoma of no special type, grade 2.

Additional Findings: Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptor (ER) Status: Positive
- Progesterone Receptor (PR) Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: Patient diagnosed with breast cancer at age 29., Sister diagnosed with triple-negative breast cancer at age 45.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 y, Family history: Sister with triple-negative breast cancer at age 45

Conclusion

The patient is eligible for NCCN testing criteria for high-penetrance breast cancer susceptibility genes.

Due to a personal history of breast cancer diagnosed at age 29 and a sister diagnosed with triple-negative breast cancer at age 45, the patient meets the criteria for genetic testing. This includes consideration of testing for BRCA1, BRCA2, among other genes mentioned in the NCCN Guidelines.

Medical Report

Pathology Report Summary

Clinical History: A 50-year-old individual diagnosed with breast cancer.

Diagnosis:

- Invasive Ductal Carcinoma: High-grade, grade 3
- Ductal Carcinoma In Situ: High grade with comedo necrosis

Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Cousin: Diagnosed with male breast cancer at age 35.
- Generation III:
- Generation IV:

Image:

NCCN Testing Criteria Assessment

Criteria: Personal history of breast cancer with age ≤ 50 years: Met, as the patient was diagnosed at age 50., Family history of male breast cancer: Met, with a cousin diagnosed with male breast cancer.

Conclusion

The patient is eligible for NCCN genetic testing based on multiple criteria. The diagnosis at age 50 and the presence of male breast cancer in a close relative strongly support the need for further genetic evaluation. Testing for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2, is recommended.

Medical Report

Pathology Report Summary

Case Number: 47

Age of Diagnosis: 50

Family History: Cousin with male breast cancer

Diagnosis: Invasive ductal carcinoma, grade 3: High-grade invasive ductal carcinoma. Ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- ER Status: Positive
- PR Status: Positive
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncles
- Generation II: Aunts, Uncles
- Generation III: The patient, diagnosed with breast cancer at age 50., Cousin with male breast cancer.
- Generation IV: Children, Nieces, Nephews

Family Link:

NCCN Testing Criteria Assessment

Criteria Met: Personal history of breast cancer with age ≤ 50 years., Any age with a close blood relative with male breast cancer.

Conclusion

The patient meets the NCCN criteria for high-penetrance breast cancer susceptibility gene testing. This includes a personal history of breast cancer diagnosed at age 50 and a close blood relative (cousin) with male breast cancer. It would be appropriate to consider testing for BRCA1, BRCA2, and other relevant genes per NCCN Guidelines. The significant family history warrants genetic consultation and potential testing.

Medical Report

Pathology Report Summary

Case Number: 48

Age of Diagnosis: 60

Diagnosis: Invasive breast carcinoma of no special type, grade 3. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Hormone Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I: Grandfather: Diagnosed with pancreatic cancer.
- Generation II: Father: Diagnosed with pancreatic cancer.
- Generation III: Patient: Diagnosed with invasive breast carcinoma, age 60, Triple-negative.

Visual Representation:

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer., Any age with ≥ 1 close blood relative with pancreatic cancer.

Conclusion

The patient is eligible for NCCN testing criteria based on having Triple-negative breast cancer and a family history of pancreatic cancer in close relatives. Testing for high-penetrance breast cancer susceptibility genes (including BRCA1, BRCA2, etc.) is recommended due to these factors.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Not specified in the document.

Diagnosis: Invasive breast carcinoma of no special type, grade 3 along with Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: Patient's Grandfather: Pancreatic cancer
- Generation II: Patient's Father: Pancreatic cancer
- Generation III: Patient: Breast cancer diagnosed at age 60, ER-negative, PR-negative, HER2-negative.
- Generation IV: Children, no noted diagnoses.

NCCN Testing Criteria Assessment

Criteria: Age at diagnosis not ≤ 50 years: The patient was diagnosed at 60., Triple-negative breast cancer: The patient is ER-negative, PR-negative, and HER2-negative, fulfilling the criteria., Family history of pancreatic cancer: The patient has multiple close relatives (father and grandfather) with pancreatic cancer.

Conclusion

The patient is eligible for NCCN genetic testing due to having triple-negative breast cancer and a family history of pancreatic cancer in close relatives. Testing may include BRCA1, BRCA2, among other genes based on the patient's clinical and family history.

Medical Report

Pathology Report Summary

Case Number: 49

Age of diagnosis: 75

Diagnosis: Invasive ductal carcinoma, grade 3 with ductal carcinoma in situ (DCIS), high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2: Positive

Pedigree Analysis

Generations:

- Generation I:
- Generation II: Aunt with ovarian cancer
- Generation III: The patient, diagnosed with high-grade invasive ductal carcinoma at age 75, ER-positive, PR-positive, HER2-positive.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria Met: Any age with ≥ 1 close blood relative with ovarian cancer.

Criteria Not Met: Personal history of breast cancer with age ≤ 50 ., Any age with Triple-negative breast cancer., Any age with Multiple primary breast cancers (synchronous or metachronous)., Any age with Lobular breast cancer with personal or family history of diffuse gastric cancer., Any age with Male breast cancer., Any age with Ashkenazi Jewish ancestry., Any age with ≥ 1 close blood relative with pancreatic cancer., Any age with ≥ 1 close blood relative with prostate cancer with metastatic, or high- or very-high-risk group., Any age with ≥ 3 diagnoses of breast and/or prostate cancer (any grade) on the same side of the family.

Conclusion

The patient meets the NCCN testing criteria due to having a close blood relative (aunt) with ovarian cancer. Therefore, the patient should be considered for testing for high-penetrance breast cancer susceptibility genes, including BRCA1 and BRCA2.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer, for review.

Specimen: Not specified.

Diagnosis: Invasive ductal carcinoma, grade 3 with ductal carcinoma in situ, high grade with comedo necrosis.

Receptor Status:

- Estrogen Receptors: Positive
- Progesterone Receptors: Positive
- HER2/neu: Positive

Pedigree Analysis

Generations:

- Generation I: Great Aunts, Great Uncle
- Generation II: Aunt with ovarian cancer
- Generation III: Affected Patient diagnosed with breast cancer at age 75.
- Generation IV: Children, Nieces, Nephews

NCCN Testing Criteria Assessment

Criteria Met: Any age with ≥ 1 close blood relative with ovarian cancer (Aunt with ovarian cancer).

Criteria Not Met: Personal history of breast cancer with age ≤ 50 years., Any age with Triple-negative breast cancer., Any age with Multiple primary breast cancers., Any age with Lobular breast cancer with personal or family history of diffuse gastric cancer., Any age with Male breast cancer., Any age with Ancestry: Ashkenazi Jewish ancestry., Any age with ≥ 1 close blood relative with breast cancer at age ≤ 50 ., Any age with ≥ 1 close blood relative with male breast cancer., Any age with ≥ 1 close blood relative with pancreatic cancer., Any age with ≥ 1 close blood relative with prostate cancer with metastatic, or high- or very-high-risk group., Any age with ≥ 3 diagnoses of breast and/or prostate cancer on the same side of the family.

Conclusion

The patient is eligible for NCCN testing based on meeting the criterion of having a close blood relative with ovarian cancer. The presence of estrogen, progesterone, and HER2/neu positive status in the pathology may also warrant additional consideration for genetic testing. Further comprehensive family history analysis and possible genetic counseling could be beneficial.

Medical Report

Pathology Report Summary

Clinical History: A case of invasive breast carcinoma, for review.

Diagnosis: Invasive breast carcinoma of no special type, grade 2. Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- Estrogen Receptors: Negative
- Progesterone Receptors: Negative
- HER2/neu: Negative

Pedigree Analysis

Generations:

- Generation I: Grandparents, Great Aunts, Great Uncle
- Generation II: Parents, Uncles, Aunts
- Generation III: Patient, diagnosed with breast cancer at age 68., Cousin with metastatic prostate cancer.
- Generation IV: Children, Nieces, Nephews

Image:

NCCN Testing Criteria Assessment

Criteria: Triple-negative breast cancer: The patient meets this criterion as both ER, PR, and HER2 are negative., Close blood relative with metastatic prostate cancer: The patient's cousin has metastatic prostate cancer, meeting this criterion.

Conclusion

The patient meets multiple NCCN criteria for testing, specifically triple-negative breast cancer and a family history of metastatic prostate cancer. Based on these findings, the patient is eligible for further genetic testing for high-penetrance breast cancer susceptibility genes.

Medical Report

Pathology Report Summary

Clinical History: A case of breast cancer for review.

Case Number: 50

Diagnosis:

- Primary Diagnosis: Invasive breast carcinoma of no special type, grade 2.
- Secondary Diagnosis: Intermediate grade ductal carcinoma in situ (DCIS) with a solid pattern.

Receptor Status:

- ER Status: Negative
- PR Status: Negative
- HER2 Status: Negative

Pedigree Analysis

Generations:

- Generation I:
- Generation II:
- Generation III: The patient, diagnosed with triple-negative invasive breast carcinoma at age 68., A cousin with metastatic prostate cancer.
- Generation IV:

NCCN Testing Criteria Assessment

Criteria Met: Any age with Triple-negative breast cancer., Any age with ≥ 1 close blood relative with prostate cancer with metastatic status.

Conclusion

The patient is eligible for NCCN genetic testing criteria based on having triple-negative breast cancer and a close familial relation with metastatic prostate cancer. Both these factors individually meet the criteria for recommending genetic testing for high-penetrance breast cancer susceptibility genes, including BRCA1, BRCA2, and others.